SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
Constituent of Symbiosis International (Deemed University), Pune
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॥वसुधैव कुटुम्बकम्॥

Department of Artificial Intelligence and Machine Learning

Course Outcomes (CO) Statements and CO-PO-PSO Mapping

Course Outcome (CO) Statements

CO1: Understand the fundamental concepts, types, applications, and data handling techniques involved in Machine Learning.

CO2: Apply classification techniques and evaluation methods for supervised machine learning problems.

CO3: Use regression techniques and performance evaluation methods for predictive modeling problems.

CO4: Use advanced classification and ensemble learning techniques for machine learning applications.

Articulation Matrix

COs \ POs/PSOs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO2
CO1	3	2	-	-	2	-	-	-	-	-	1	3	2
CO2	3	3	2	2	3	-	-	-	-	-	1	3	2
CO3	3	3	2	2	3	-	-	-	-	-	1	3	2
CO4	3	3	3	2	3	-	-	-	-	-	2	3	3

CO-PO-PSO Justification

CO	PO/PSO	Level	Justification
CO1	PO1	3	Applies mathematics, computing, and engineering fundamentals to understand ML concepts and data preprocessing techniques.
	PO2	2	Analyzes datasets and identifies suitable ML approaches for different problem domains..
	PO5	2	Uses programming tools, ML libraries, and data handling techniques for preprocessing and analysis.
	PO11	1	Develops foundational knowledge necessary for continuous learning in AI/ML technologies.
	PSO1	3	Provides the basic understanding required to analyze and design AI/ML solutions.
	PSO2	2	Introduces ML concepts that support advanced AI domains and platforms.
CO2	PO1	3	Utilizes mathematical and computational principles behind classification algorithms.
	PO2	3	Formulates and analyzes classification problems to select appropriate algorithms.
	PO3	2	Develops ML-based solutions for classification tasks in real-world applications.
	PO4	2	Conducts experiments and evaluates classifier performance using metrics such as accuracy, precision, recall, and F1-score.
	PO5	3	Employs ML tools and software platforms to implement classification models.
	PO11	1	Encourages learning of emerging classification methods and evaluation strategies.
	PSO1	3	Applies AI/ML techniques to solve practical classification problems.
	PSO2	2	Supports implementation of intelligent systems in domains such as NLP and Computer Vision.
	PO1	3	Applies mathematical and statistical foundations of regression algorithms.
	PO2	3	Analyzes predictive modeling problems and selects suitable regression techniques..
	PO3	2	Designs predictive solutions using regression models.

CO3	PO4	2	Conducts experiments and interprets regression performance measures such as RMSE, MAE, and R^2.
	PO5	3	Uses modern ML tools and software for regression model development.
	PO11	1	Enhances capability to adapt to evolving predictive analytics technologies.
	PSO1	3	Enables development of AI/ML-based predictive systems.
	PSO2	2	Supports advanced applications involving forecasting and intelligent decision-making.
CO4	PO1	3	Applies advanced ML knowledge and computational techniques.
	PO2	3	Analyzes complex datasets and selects suitable ensemble learning approaches.
	PO3	3	Designs robust ML solutions using ensemble methods such as Random Forest, Bagging, and Boosting.
	PO4	2	Investigates and compares ensemble models through experimentation and performance analysis.
	PO5	3	Utilizes advanced ML tools and frameworks for implementation.
	PO11	3	Encourages exploration of emerging ensembles and advanced learning techniques.
	PSO1	3	Develops intelligent systems using advanced AI/ML methodologies.
	PSO2	3	Applies specialized ML techniques relevant to AI domains and industrial applications.